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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 39: Circular Economy in Transport Sector

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Need of CE in transport sector

- With economic growth & development, mobility systems are under increased pressure and without a change of approach, will result in negative externalities.
- Transport sector a key contributor in emissions, resource consumption.
- Adoption of CE concept will support the overall economic, environmental, and social prosperity of the city.

Need of CE in Urban mobility system

- Mobility is a key urban priority
- It is central to how a city operates, and has a significant impact on quality of life, the local environment, and resource consumption.
- Effective urban mobility systems can be enabled by accommodating all modes of transport.
- Adoption of CE concept will optimize the transport sector by integrating mobility planning with the spatial planning while improving products and services.

CE adoption in Vehicle's manufacturing

- Enabling repair, reuse, remanufacturing, and recycling of components and materials.
- Need to consider how vehicles are designed so that we can easily repair them during service and recover and reuse their materials at end of use.
- A lot of energy is used to create vehicle parts, and that energy is not wasted if the parts - or materials - are reused.

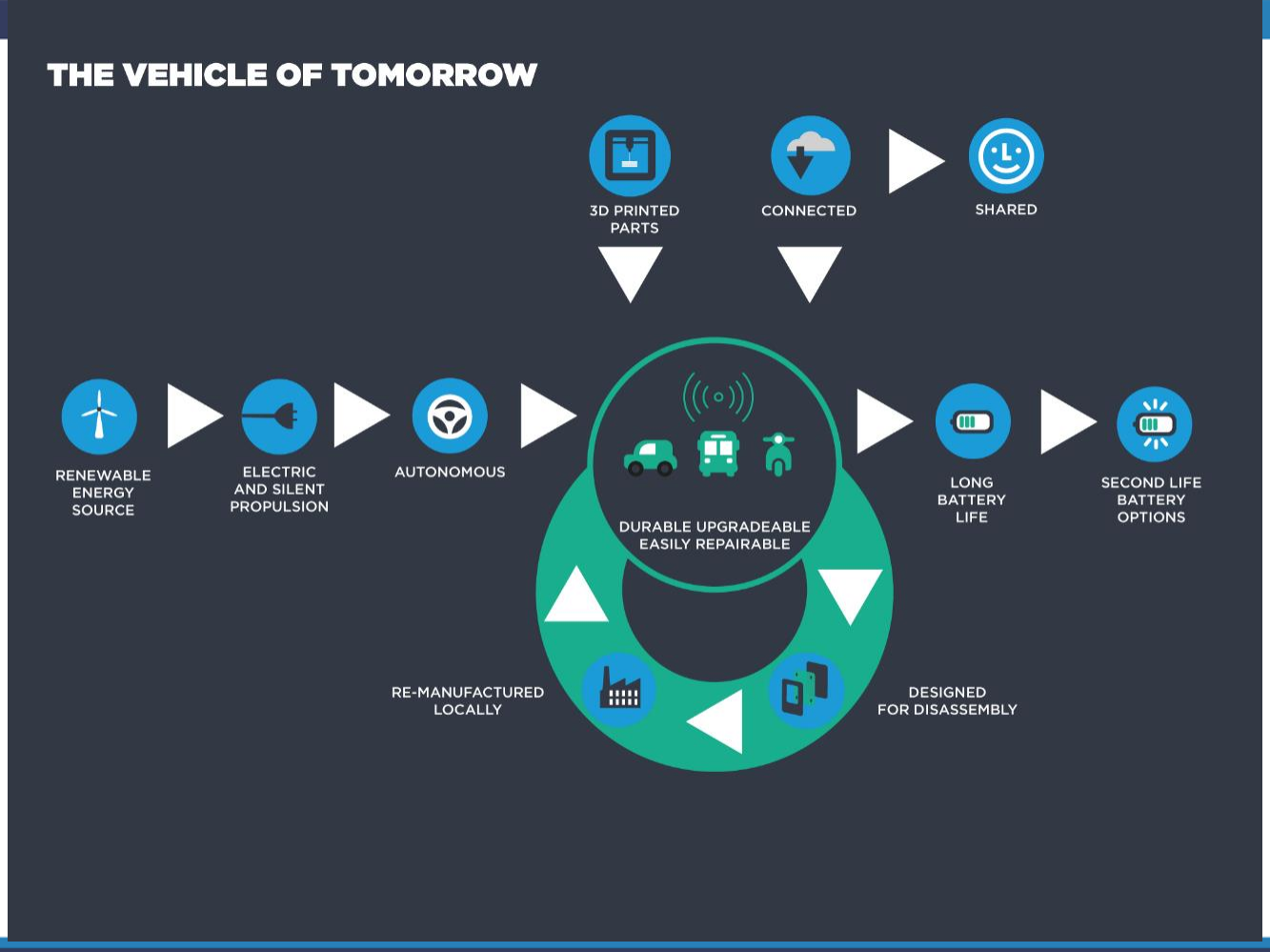
Strategies for vehicle manufacturing in CE

- Designing vehicles for adaptable and shared use
- Maximum use potential and the ability to remanufacture
- Modularity, in which parts can be swapped in and out, and/or space in the vehicle can be reconfigured, can enable longer use while responding to the user's needs for changes and repair
- Flexible interiors in freight vehicles can be designed to support the simultaneous delivery and collection of goods and other uses as per needs

Example of CE in vehicle manufacturing

- Car manufacturer, Renault adopted CE concept
- Variety of materials including copper, steel, textiles, and plastics taken from ‘end-of-life vehicles’ (ELV) and put into new vehicles with the same level of performance as virgin materials
- Currently, 36% of the total mass of a newly produced Renault vehicle in Europe is made from recycled materials, and 85% of an ELV is recyclable.





Adoption of CE in vehicle manufacturing process

Adoption of CE in transport infrastructure

- Designing transport infrastructure for adaptable use

- Usage of construction material which have minimum negative impacts and material can be reuse after completing its life

- Dynamic design of infrastructure

Example: Dynamic street-use restrictions which promote specific transport modes at certain times of day (prioritizing road space for bikes indicated by LED lights in the road, or freight-lane restrictions).

- Simple and low-cost design interventions provided such as moveable street furniture

Adoption of CE in mobility planning

- Compact city development for effective mobility

Develop infrastructure in a way which promote environment friendly mobility such as walking, bicycling

TOD is also a part of CE concept adoption in mobility planning for urban areas

Example: Transit orientated development in cities or walking friendly infrastructure

Example: London Walking Plan 1/2

- By 2030, congestion is projected to cost London GBP 9.3 billion a year.

- To counter this, the city is rolling out an extensive transport plan, which includes a dedicated walking action plan aimed at adding 1 million additional walking trips a day.



Image source: intelligenttransport.com

Example: London Walking Plan 2/2

- The city will invest GBP 2.2 billion to redesign streets, install better signposting and maps, add more pedestrian crossings, as well as improve public transport.

- Analysis shows that a walkable London is significantly more land-use efficient, can save up to GBP 1.6 billion in public healthcare costs, increase retail sales, reduce emissions, and increase social cohesion and living conditions

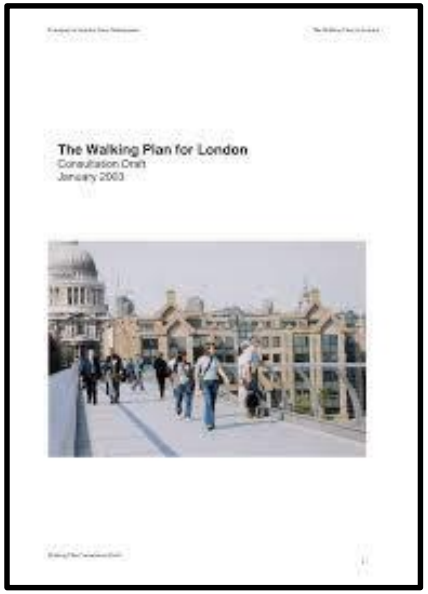


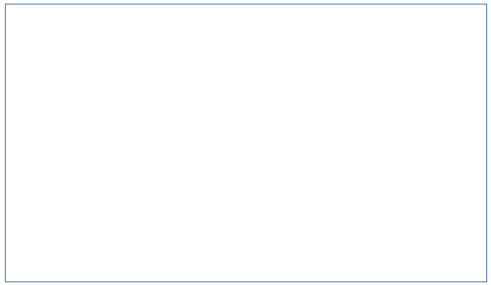
Image source: London.gov.uk

Adoption of CE in mobility planning

- Urban freight strategies for effective reverse

Planning the freight movement in an optimized manner

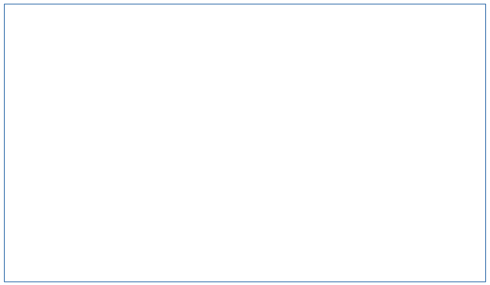
Coordination between various delivery/logistic companies so that resources used are minimum and it will also reduce the emission value for per kg logistic transfer from one place to another.



Example: Paris' logistics hotels and inner-city green freight 1/2

- Paris is bringing logistics facilities back into the city with the use of so-called 'logistics hotels' in high-density areas.

- These urban hotels help to reduce heavy vehicle use and the associated negative impacts, such as emissions and noise, while increasing the productivity of the delivery services.



Example: Paris' logistics hotels and inner-city green freight 2/2

- Parcels from suburban logistics centers are pooled at the 'logistics hotels' via freight train services or a limited number of larger delivery shuttle-trucks.

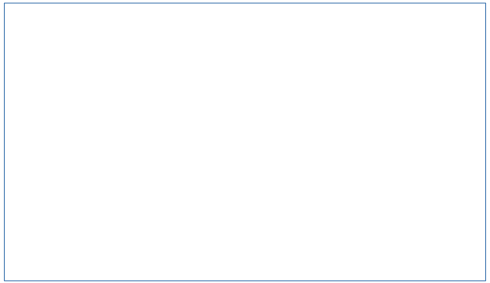
- From there deliveries are made in smaller, lower-emission vehicles such as electric tricycles that can carry up to 180 kg. The space in these 'logistics hotels' is rented out by the city at a favorable rate, but requires delivery firms to use low-emission transport modes

Adoption of CE in mobility planning

- Resource flows (capital & material)

Designing and planning of mobility in a way which will minimize the resource utilization in terms of capital and also resources.

A public transport system may carry greater number of people at a time with less investment and lesser emissions



Example: Potential benefits of CE in transport sector of China

- Circular economy opportunities could reduce the total cost of access to mobility by CNY 12.6 trillion (USD 2 trillion) in 2030: an improvement of 36% over the current development path

- Implementing circular economy opportunities would result in a decrease in consumption of non-renewable resources, including fossil fuels, by 49% in 2030 and 71% in 2040

Source: Ellen MacArthur Foundation, The Circular Economy Opportunity for Urban and Industrial Innovation in China (2018)

Example: Reversing car-centric mobility in Sao Paulo

- The city's 2014 urban masterplan includes a focus on the urban mobility system to expand public and active transport modes.
- The intention is for the measures in the plan to help increase the number of residents living near public transport from 25% in 2015 to 70% in 2025.
- The city allocated 30% of urban development funds towards this effort.
- The masterplan aims to unlock economic, societal, and environmental opportunities including supporting the provision of more affordable housing and improving economic opportunities for urban residents

Mobility system under CE

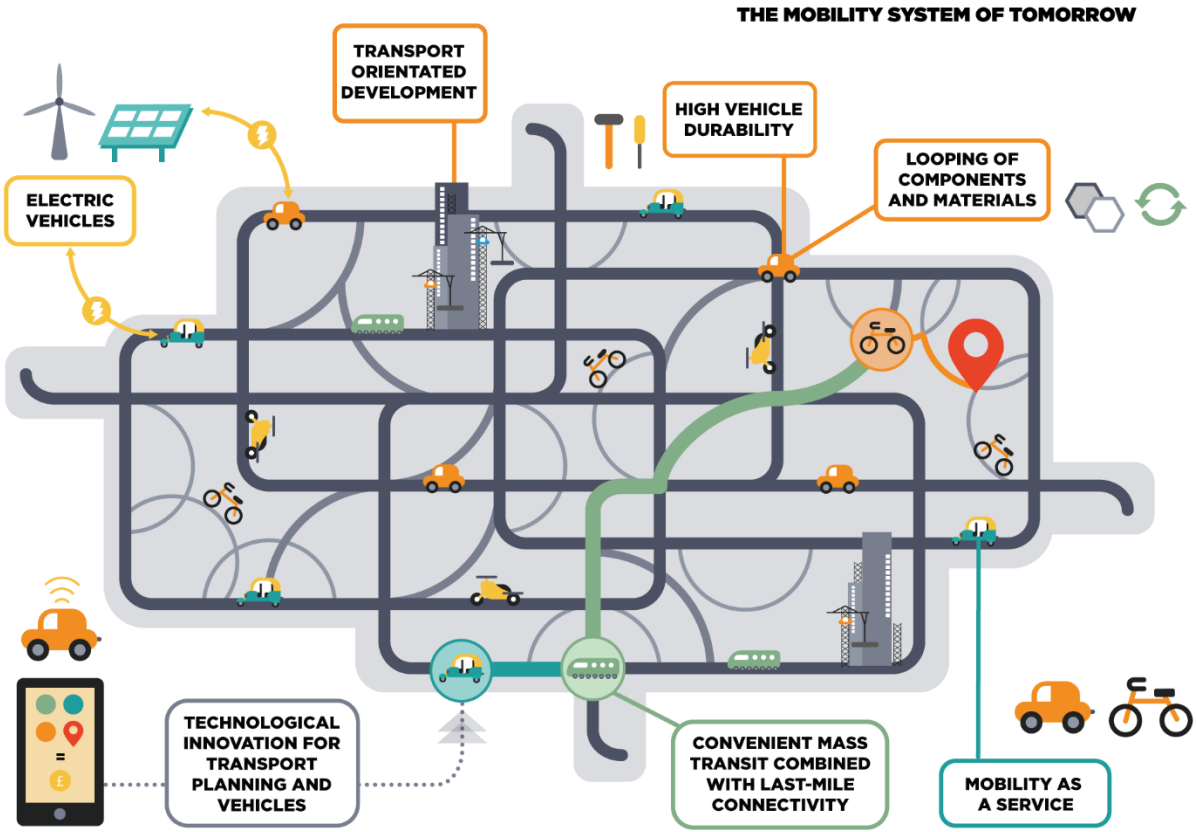
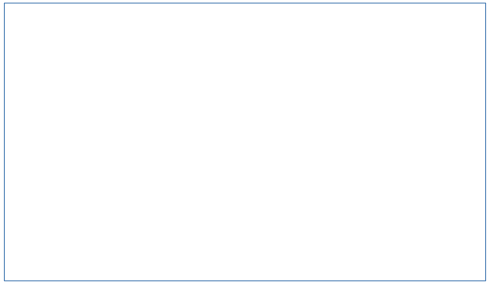


Image shows an urban mobility system after adopting CE concept which promote walking, public transport etc.

Adoption of CE in transport Policy & regulations

- Policies, regulations and incentives can be a strong tool for promoting CE concepts in transport sector.
- Promoting recycling of vehicles or incentive on vehicle scrappage, high standards for vehicle manufacturing can ensure less pollution and less resource consumption.
- Example: Scrappage policy of India adopted in year 2021 is an example.



Vehicle Scrappage policy of India

- Policy introduced in annual union budget 2021-22
- Formal regulations are announced recently (August 2021).
- Ensure the scientific methods to scrap the used cars.
- It will promote material reuse.
- Timely scrappage consists incentive like no registration fee on purchase of new vehicle which will keep polluting old vehicles out of system.

Scrape value of old vehicles will be around 4-6% of ex-showroom price of new vehicles which will be beneficial for users.

It will promote transfer from BS-IV to BS-VI vehicles in a rapid manner.

Case Study:

Creating a Circular City–An analysis of potential transportation, energy and food solutions in a case district

- Partial case study has been used as example.
- Transport sector related results and data has been taken.
- Impacts of adoption of CE is discussed.

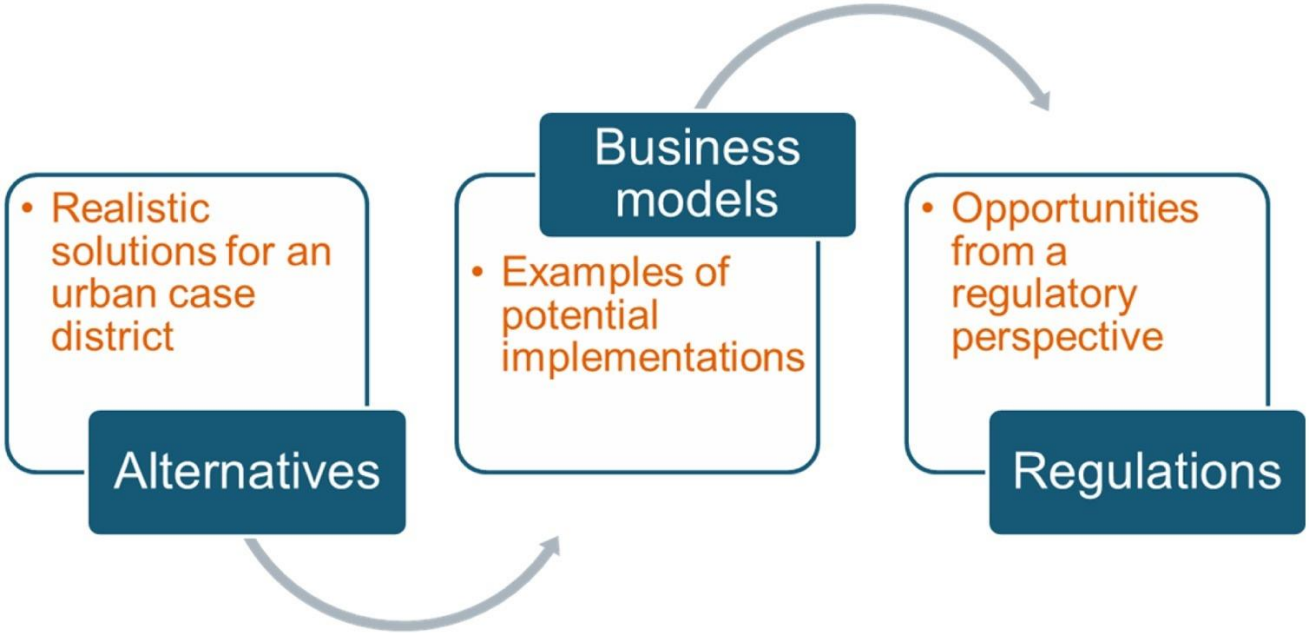
Study Area: Kera, Espoo, Finland

- All the analyses were performed in Kera, which is a new planned housing area in the city of Espoo, Finland.
- Kera was selected as a case district because it is a typical example of where a former industrial area is rezoned into a housing area with circular city targets.
- The strategical target is that Kera is going to be an international example of circular economy principles implemented at the city level



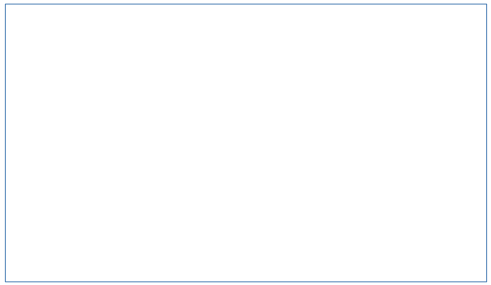
Image source: bm-ark.fi

Approach of the study



Objectives of CE in Kera, Espoo 1/2

- Carbon neutrality by 2030
- Achieving UN Sustainable Development Goals by 2025
- Emission-free public transport by 2030
- Enabling MaaS (Mobility as a Service) for residents



Objectives of CE in Kera, Espoo 2/2

- Deployment of electric buses
- Completely renewable energy production by 2030
- Emission-free district heating by 2030
- Effective utilization of waste as material and energy
- Promoting the utilization of surplus food

General attributes of study area

	District of Kera	City of Espoo	Unit
General data			
Total area	79	52,800	ha
Inhabitants	13,300	289,731	–
Jobs	10,000	120,676	–
Buildings			
Residential floor area	532,000	10.2 million	m^2
Business, service etc.	147,000	10.6 million	m^2
Energy demand			
Annual heating demand	43		GWh/year
Annual cooling demand	4		GWh/year
Annual electricity demand	30		GWh/year
Transportation			
Number of (person) trips	115,693		–
Average trip length	12.60		km/trip

Scenarios for Transport system

Scenario 1 Conservative, relying on combustion engine technologies	Deduced based on the year 2035 figures for the whole of Finland
Scenario 2 Accelerated adoption of carbon-free technologies	It is assumed that the penetration of new technologies is faster in the largest cities (such as Espoo) because • Distances are shorter than in more sparsely populated areas • Alternative refueling and charging networks will develop favorably
Scenario 3 Ambitious future vision (CE concept adopted)	Public transport should be carbon-free by the year 2030. Additionally, this scenario assumes that the inhabitants of Kera are early adopters resulting in all hybrid or electric passenger car base.

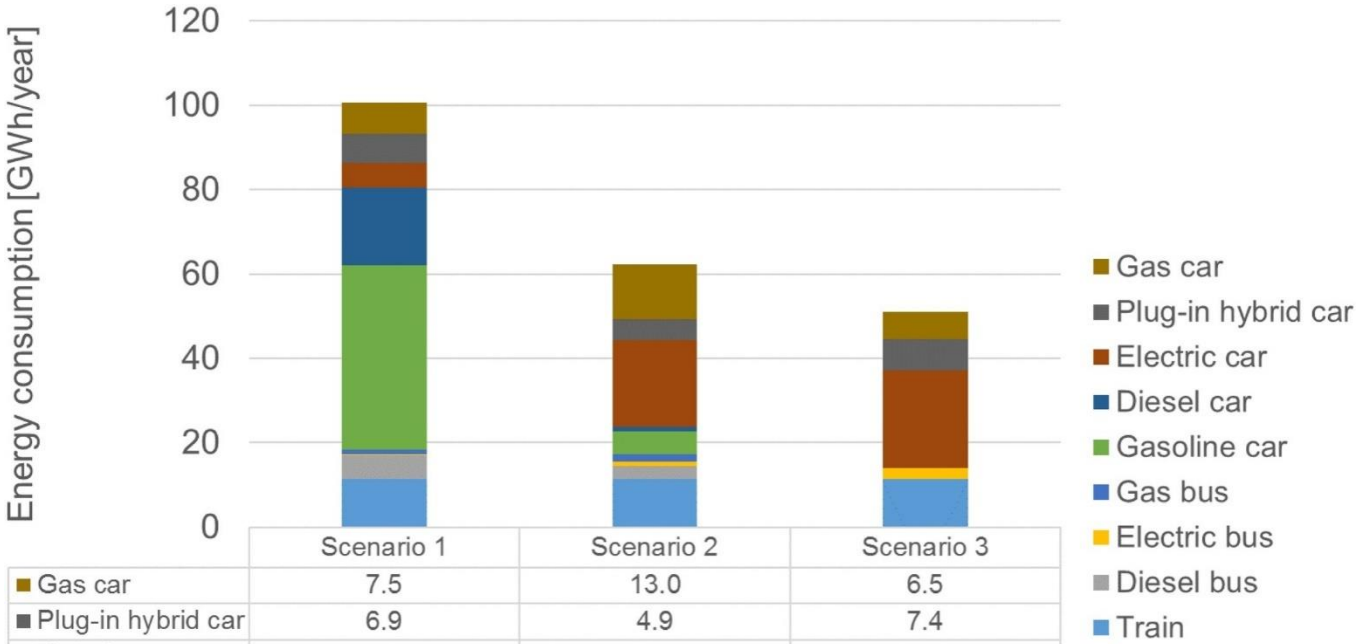
Transportation scenarios for the Kera area in 2035

Travel mode			Technology	Scenario 1	Scenario 2	Scenario 3
Public transport	**Train**	70 %	Electric	100 %	100 %	100 %
			Diesel	72 %	40 %	0 %
	Bus	**30 %**	**Electric**	15 %	40 %	100 %
			Gas	13 %	20 %	0 %
			Gasoline	**40 %**	5 %	0 %
Passenger car			**Diesel**	18 %	1 %	0 %
			Electric	20 %	70 %	78 %
			Plug-in hybrid	14 %	10 %	15 %
			Gas	8 %	14 %	7 %

Energy consumption & emission values for vehicles

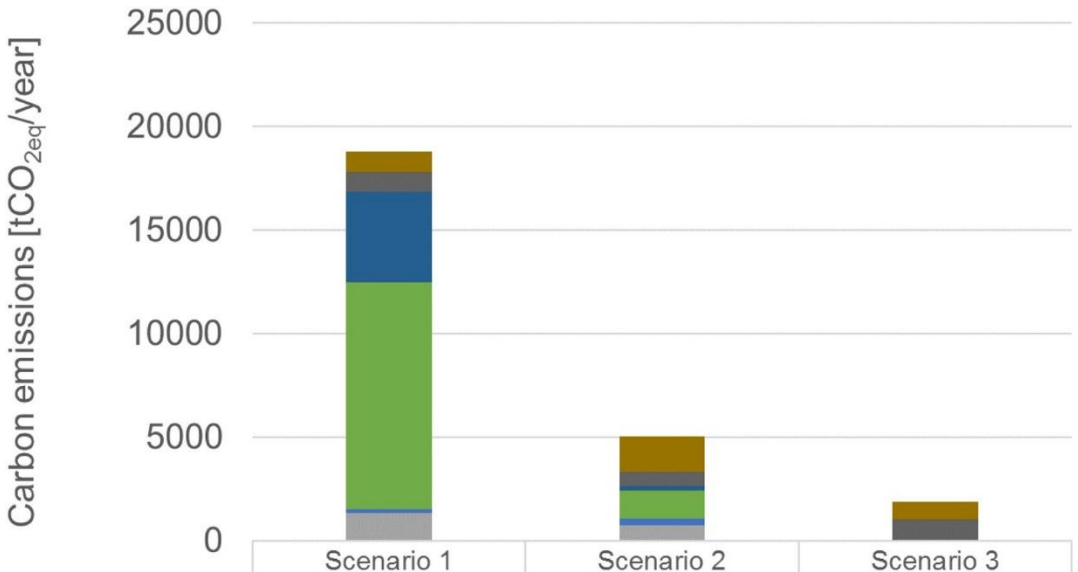
Vehicle type	Vehicle powertrain	Energy consumption [kWh/pkm]	CO_{2eq} emissions [g/pkm]
Passenger cars	**Gasoline**	0.37	93.61
	Diesel	0.35	82.93
	Gas	0.32	41.25
	Plug-in hybrid	0.17	23.40
	Electric	0.10	0.00
	Diesel	0.17	41.35
Buses	**Gas**	0.19	35.37
	Electric	0.06	0.00
Train	Electric	0.08	0.00

Annual energy consumption of transportation in Kera in 2035



	Scenario 1	Scenario 2	Scenario 3
Gas car	7.5	13.0	6.5
Plug-in hybrid car	6.9	4.9	7.4
Electric car	5.9	20.7	23.1
Diesel car	18.3	1.0	0.0
Gasoline car	43.7	5.5	0.0
Gas bus	1.1	1.7	0.0
Electric bus	0.4	1.0	2.6
Diesel bus	5.6	3.1	0.0
Train	11.4	11.4	11.4

Annual carbon emissions from transportation in Kera in 2035



	Scenario 1	Scenario 2	Scenario 3
Gas car	965.5	1689.6	844.8
Plug-in hybrid car	958.6	684.7	1027.1
Electric car	0.0	0.0	0.0
Diesel car	4367.4	242.6	0.0
Gasoline car	10955.4	1369.4	0.0
Gas bus	204.8	315.1	0.0
Electric bus	0.0	0.0	0.0
Diesel bus	1326.1	736.7	0.0
Train	0.0	0.0	0.0

Conclusions of the Kera study

- In the best case (scenario 3), the energy consumption of the transportation in Kera area could be halved and second, the emissions decreased to 10 % of the base level (Scenario 1) through the introduction of cleaner technologies, mainly via electrification.

- Best scenario depends on various factors, e.g., on the availability and costs of the cleaner, more energy efficient alternatives but also on the measures and incentives implemented

- Significant share of the gas-powered vehicle fuel could be produced locally using waste as energy resources.

Final Conclusions

- In transport sector, CE concepts can be adopted in mobility planning, vehicle manufacturing and infrastructure development.
- Adoption at each step can give vey positive impacts in terms of resource/energy consumption and vehicular emissions.
- Various approaches such as TOD, adoption of EVs, walking & bicycling friendly infrastructure are lead us to a circular system.
- Using waste as energy resource for vehicles, recycling of vehicles are should also be adopted for CE.

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Thank You